

Firewalls and IDS/IPS — Guided Notes

Unit 4.2.1 | CompTIA Network+ N10-009 Obj. 4.3 | N10-008 4.3

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A stateless firewall evaluates each packet _____, while a stateful firewall maintains a _____ table that tracks active connections. Why does stateless filtering require explicit rules for return traffic?

Question 2

A next-generation firewall (NGFW) differs from a traditional stateful firewall by inspecting the _____ layer, allowing policy decisions based on _____ identity rather than just port number.

Question 3

Firewalls process rules _____. If a broad deny rule appears above a specific permit rule, the permit rule will _____ be reached. Best practice is to place _____ rules at the top.

Question 4

An implicit deny is a rule at the _____ of the firewall rule set that blocks all traffic not matched by a preceding rule. Why is it important to _____ this rule?

Question 5

A DMZ (demilitarized zone) places public-facing servers on a _____ network segment between the internet and the internal network. If a DMZ web server is compromised, what prevents the attacker from immediately reaching internal systems?

Question 6

An IDS is _____ — it receives a copy of traffic and generates alerts. An IPS is _____ — traffic passes through it and it can actively _____ malicious packets. What happens to network traffic if an inline IPS fails in fail-closed mode?

Question 7

Signature-based detection matches traffic against _____ attack patterns, so it cannot detect _____. Anomaly-based detection flags deviations from a _____ but produces more _____ than signature-based systems.

Question 8 — Real World Application

REAL WORLD: A network admin adds a firewall rule permitting guest VLAN traffic to the student portal on port 443, but the portal remains unreachable from the guest network. The rule is syntactically correct. What is the most likely cause, and how do you verify it?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/4-2-1-firewalls-and-idsips/4-2-1-firewalls-and-idsips-answer-key.pdf
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